Fiscal Year:	FY 2018	**Task Last Updated:**	FY 05/13/2021
PI Name:	Diddams, Scott A. Ph.D.		
Project Title:	Development of an Optical Microcomb System in Support of Fundamental Physics Research on the International Space Station		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Boulder	**State:**	CO
Zip Code:	80305-3337	**Congressional District:**	2
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2011 Fundamental Physics NNH11ZTT001N
Start Date:	06/01/2012	**End Date:**	05/31/2018
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JPL
Contact Monitor:	Israelsson, Ulf	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: Extended to 5/31/2018 per PI (Ed., 3/7/17)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Papp, Scott (National Institute of Standards & Technology) Vahala, Kerry (California Institute of Technology)		
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Performance Goal Text:			

Task Description:	Our research program will investigate and develop advanced microresonator optical frequency comb technology in support of planned European Space Agency (ESA) activities on the International Space Station (ISS). Optical frequency combs, which provide a dense set of absolute frequency reference lines, are an integral component of modern optical atomic clocks. They provide a link between optical and microwave signals and a means to access the stability of optical clocks at any wavelength. For example, a frequency comb will be required for the Space Optical Clock (SOC) experiment to achieve all of its fundamental scientific goals by providing the critical phase coherent optical or microwave link between the atomic frequency standard on the ISS and ground-based clocks. The same frequency comb technology will be a required component of other proposed ESA and NASA fundamental-physics clock experiments, as well as an enabling technology for ESA's planned Quantum Weak Equivalence Principle (QWEP) atom interferometry experiments. Microresonator combs (microcombs) can be lithographically patterned and fabricated on a single silicon chip with dimensions of a few square centimeters and therefore offer revolutionary advantages beyond any other conventional laser comb technology in terms of size, power consumption, and sensitivity to their environment. However, the full capability of a microcomb for precision optical frequency metrology has not been realized. The laboratory experiments we propose will, for the first time, characterize a microcomb's stability with respect to known optical standards, thus verifying its suitability to support the high-performance optical clocks envisioned for SOC and other space missions. Our microcomb system will be tested and interfaced with state-of-the-art optical clocks based on strontium and ytterbium atoms, which are the two candidate SOC platforms. Furthermore, the microcomb will have an output at 1064 nm and a countable microwave repetition rate in the 10-50 GHz range, which are compatible with the optical and microwave links required to achieve the fundamental physics goals of the SOC mission. Our research will develop advanced technology to enable full on-chip integration of a microcomb with essential components for time and frequency metrology including high-Q silica microresonators, clock-signal coupling, comb self-referencing, and fiber v-groove end-couplers. These efforts will contribute to the long-term goal of realizing a single chip-scale device operating at the convergence of optics, frequency metrology, opto-mechanics, and microwave technology.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The best optical clocks now keep time with uncertainty measured in the 18th digit; however, their complexity constrains such remarkable precision to a research laboratory setting. Our research is focused on developing key components of optical clocks that will eventually enable their use outside the lab and in outer space. In particular, we aim to develop the compact “clockwork” that will enable the counting of optical cycles on the femtosecond time scale, as well as the synthesis of optical and microwave frequencies critical for transmission and comparison of multiple atomic clocks. Precision time keeping, as explored in our research, is anticipated to impact a wide variety of human activities both in Earth and space. Not only do clocks provide an accurate tool to analyze the inner workings of atoms and molecules, but they are critical to communication and navigation systems that we rely on every day. Examples include the global positioning system and cellular phone networks. Our research is an important step in developing the next generation clocks which will not only be more accurate, but also more compact and easily fabricated.
Task Progress:	In this period of the program, our research activities were focused on: Self-referencing of soliton micro-resonator frequency combs. Our effort has aimed to streamline the self-referencing of both single and double solitons from a 15-GHz silica disk resonator using traditional f-2f techniques. Since this work relies on directly generating a single or double soliton state with no additional waveform shaping, a key component of our experiment is to reproducibly generate single solitons and maintain them through active feedback, while providing frequency control of the comb spectrum for the repetition rate and carrier-offset frequency stabilization. The active soliton feedback is implemented by monitoring the optical power after the resonator and adjusting the pump laser frequency in order to hold this power level constant, which can keep the soliton pulses stable for more than a day. Then, amplification in an erbium-doped fiber amplifier (EDFA) and pulse compression in normal-dispersion highly nonlinear fiber (HNLF) is utilized to achieve pulses with energies up to 180 pJ and durations of 100 to 120 fs. These pulses are used to generate coherent supercontinuua in silicon-nitride wave- guides, which have recently been demonstrated as suitable platforms for allowing frequency-comb self-referencing and precision metrology with low-pulse-energy lasers. This use of silicon-chip-based devices for both the soliton generation and octave-plus broadening shows the potential for chip-integrated sources of broadband, high- repetition-rate frequency combs at the 15-GHz repetition rate, which is useful for its ability to resolve individual comb lines while remaining detectable with microwave electronics. Additionally, using a two-soliton state from the resonator in conjunction with the silicon-nitride waveguide, we show a fully self-referenced comb using standard f-2f interferometry and use the comb to compare two cavity- stabilized lasers separated by 87 THz across the near infrared. Our work outlines and demonstrates the nonlinear optics and frequency-metrology pathway for a self- referenced, chip-integrated microcomb system, provided that 100 pJ, < 100 fs soliton pulses are derived from the Kerr microresonator and subsequent optical amplification. Further chip integration is possible in the future by implementing the pulse compression stage in silicon-nitride waveguides.
Bibliography Type:	Description: (Last Updated: 05/13/2021)
Articles in Peer-reviewed Journals	Cole DC, Lamb ES, Del'Haye P, Diddams SA, Papp SB. "Soliton crystals in Kerr resonators." Nature Photonics. 2017 Oct;11:671-6. https://doi.org/10.1038/s41566-017-0009-z , Oct-2017
Articles in Peer-reviewed Journals	Lamb ES, Carlson DR, Hickstein DD, Stone JR, Diddams SA, Papp SB. "Optical-frequency measurements with a Kerr-microcomb and photonic-chip supercontinuum." Phys Rev Applied. 2018 Feb;9(2):024030. https://doi.org/10.1103/PhysRevApplied.9.024030 , Feb-2018
Articles in Peer-reviewed Journals	Cole DC, Papp SB, Diddams SA. "Downsampling of optical frequency combs." J Opt Soc Am B. 2018 Jul;35(7):1666-73. https://doi.org/10.1364/JOSAB.35.001666 , Jul-2018
Articles in Peer-reviewed Journals	Stern L, Stone JR, Kang S, Cole DC, Suh MG, Fredrick C, Newman Z, Vahala K, Kitching J, Diddams SA, Papp SB. "Direct Kerr frequency comb atomic spectroscopy and stabilization." Sci Adv. 2020 Feb 28;6(9):eaax6230. https://doi.org/10.1126/sciadv.aax6230 ; PMID: 32158936; PMCID: PMC7048413 , Feb-2020